

Energy sector reforms — I

By Sheikh Imran ul Haque

Business Recorder · Opinion · Published 29 December 2022

My previous op-ed, summed up Pakistan's gas supply position vis-à-vis the current demand, prevailing geo-political context, international spot-market under which Pakistan functioned over the last 4 years and the much criticized "Qatar LNG gas deal" which if not signed, the economy today would have potentially grind to a halt due to severe gas shortage, extensive rolling blackouts, contraction and shutdown of industry to make up the 600 mmscfd contracted in 2016/17. Warm winter is ending.

Additionally, there are factors likely to persist among which, a finite set of molecules in the global fuel-supply chain is likely to dominate energy markets. The implications of a price cap of USD 60 on Russian oil are yet to be understood, Japan Korea Method (JKM) of over USD 38.6 for February contracts means continued underutilization of Port Qasim terminals. The long-term deals of Sinopec for 27 years and Germany for 15 years starting 2025/26 with Qatar point towards action pending from Pakistan and possible discounted purchase of crude and refined products from Russia.

The silver lining and the savior continues to be LNG with RLNG costing SSGC \$14.85/SNGPL \$14.47 per mmbtu at Brent of \$91.62 in Nov 2022 vs Nov 2021 of SSGCL \$15.43/SNGPL \$15.68 per mmbtu at Brent of \$76.34. (OGRA). And as per Minister of State, Musaddiq Malik, Iran would donate liquefied petroleum gas (LPG) worth £2 million to Pakistan on humanitarian basis. How long can we continue with such dole outs including living on expectation of increased deferred payment for oil?

Current, past, and future scenarios call for us to improve our energy management, reduce inefficiencies, implement strong energy conservation measures, and align long-term policies for sustainability and affordability as well as economic security – which cannot be seen outside the lens of energy security. Driving these policies require that we fix the head for the body to follow as the future will always be unpredictable. And to fix the head, hard realities require acknowledgement with approved policies that have a strict commitment to transition the sector to stability with agile operations and decision taking.

Our past has shown that historically, government of Pakistan (GOP) is unable to comprehend the task as aptly advocated by Yousuf Nazar – an independent investment banker – that Pakistan's biggest problem is intellectual deficit. Intellect is a handicap in the intellectual wasteland. Tabish Gauhar had earlier identified the need to induct experienced and knowledgeable resources into ministry of energy (MoE) which is a conversation for another day.

On the fiscal front, Pakistan should build its revenue base and not reduce fuel retail prices (Brent \$82.02 up from \$77.1). PDL increase on HOBC is useful and 17% GST needs to be imposed on fuels.

Some immediate measures to tame the burgeoning circular debt within the gas sector requires releasing PSO's receivables from the gas utilities to restore the balance sheets of SOEs. Receivables have crossed PKR 1 trillion with experts suggesting it will soon match levels with the power sector. This requires a prudent but necessary call to action to reduce the burden on the exchequer including deploying strict energy conservation measures.

Currently, rationing of gas for 16 hours a day is in place; SSGC and SNGPL seek a tariff hike of Rs 1,360/Rs 1,840 per mmbtu from Rs 693/Rs 294 with local gas priced at \$4.4-5.5 per mmbtu. The gas sector circular debt has increased to Rs 1,900 billion and yet GOP continues to offer export and processing firms \$10 per mmbtu. For RLNG diverted to domestic sector, MoE requires Rs 105 billion in next 4 months in addition to Rs 174bn overdue for last 4 winter seasons. (Newspapers, Twitter).

Fiscal slippages are resulting in calls to mobilise an additional Rs 800 billion (1% of GDP) revenue as per news reports. APTMA's Rs 100 billion fuel subsidy would actually consume Rs 200 billion as per the IMF, the impact of the Kissan Package on the budget at around Rs 180 billion and doling out Rs 536 billion for export-oriented industry when in return there was hardly \$2 billion increase in exports — due to price increases, not due to quantity.

The message is clear: We cannot afford unfunded subsidies in the short term. We need to charge the cost of fuel, increase tariff and fund targeted subsidies that can be delivered directly to consumers. A wiser decision is to throttle down on wholesale subsidies and divert such funding to low-income consumers to weather the economic stagnation while streamlining efficiency standards. We are at that stage where gas supply rates will have to be raised, stepwise, as was done for the power sector at the tail end of the last fiscal year and strengthen the legal framework surrounding the recovery of gas bills.

Already, households use 2.5 times more gas than electricity possibly due to the unfunded gas subsidy. This is fiscally unsustainable given our current gas shortfall which hovers between 700-800 mmscfd (equivalent LPG of 15-17,000 tpd) and at current trends, imported gas to meet today's demand in 2032 will be approximately 50 LNG cargoes a month or 5,000 mmscfd thru IP, TAPI and coal gasification (whose earlier studies have not resulted in conclusive cost effective approach) to only meet current and not future requirements which requires a percentage increase, double of GDP growth.

It's a staggering forecast accounting for indigenous gas reserves running out in 10 years, energy demands for 270 million citizens (presently 230 million) and given we produce 14.6% of our crude, 70% of our gas, 41.5% of our LPG, 34% of our coal and 47% of finished products while the balance is part of our import bill according to Petroleum Institute of Pakistan and Petroleum Club of Pakistan (PIP and PCP). Long term options are not mutually exclusive; rather a combination thereof can only deliver. Household consumers cannot be burdened to generate energy demand. Our focus must be on industrialization, productivity and energy conservation measures. Dependency and advocacy on poor man's fuel is a surprise instead of exploration of offshore areas (Dr. Qamar J. Sharif) and E&P investment in the UAE and Reko Diq.

Subsidies also do not justify digital warriors who think of renewables (RE), hydrogen, EV or battery storage without improving our current way of operations and charging actual fuel cost. Thus, a medium-term priority and mass production of batteries, EVs and PVs requires ease of business supported by EDB and Ministry of Science & Technology. Sanctity of commercial contracts and judiciary intervention in executive functions needs to end.

Renewable Energy (RE) sources are cheaper than thermal power in terms of fuel inputs and their technologies are increasing efficiency and reducing cost (solar and wind); however, their intermittent nature can be complemented with hydel and thermal. Their higher CAPEX due to higher requirements for backup power to meet the gap due to intermittency and associated grid challenges are a technical constraint. It is estimated at 1 MW of intermittent RE requires at least 0.9 MW of fossil-fuel inputs, ideally Gas Turbine (CCGTs) as RE is just an add-on and runs on shoulders of fossil-fuels or nuclear. Germany after spending half a trillion dollars on RE, still maintains 89% of its fossils.

Optimally availing non-intermittent and cleaner energy fuel sources is one way to fill the gap. Energy expert and one of the proponents of the Glare Initiative, Munawer Hassan, has been promoting WAPDA to take lead as Water and Alternate Power Development Authority for best utilization of its infrastructure at its hydel sites and this is an excellent concept. Farzana Yaqoob hits the nail by stating that "everyone is looking out of the box not realizing that the box has never been explored". We must make do and use and find solutions with what we have. Confidence building necessitates closure on hearings on National Accountability Ordinance (NAO) amendments (25 to-date) which is expected to continue into the new year.

Thus, available opportunities must be exploited over the next 10 years and the energy sector be forced to step out of its protected comfort zone through deregulation as we are running out of options despite the availability of an improved energy mix.

Pakistan will always be in a catching up mode to meet its energy demands despite best efforts to reduce the power sector circular debt down by 250bn to Rs 2217 bn. This means significantly reducing Transmission and Distribution (T&D) and Unaccounted For Gas (UFG) losses, improving DISCO collections while driving up system efficiencies and energy conservation standards.

In the power sector these losses stand at 17.13%. For 4 bscf consumption in Pakistan, distribution losses are 13.86% and 14.14% for SNGPL and SSGC, respectively: Collectively, the two represent 560 mmscfd as lost units. This repeatedly justifies capex to reduce T&D and UFG thru investment by the utilities limited due cash shortfall.

Loss reduction in the gas and power sector is a no brainer given losses of 600 mmscfd (USD 3-7 billion pa at USD 30/mmBtu LNG) and up to 14,000 MWs (approx. USD 14 billion investment in generation capacity excluding yearly OPEX and capacity charges based on the 2031 IGCEP Installed Capacity planning). Balance sheet of Discos and SSGC/SNGPL is limiting despite economics strongly justifying investment needed to reduce T&D and UFG urgently for which regulator's enforcement needs effectiveness.

Sheikh Imran ul Haque

The writer has served as Managing Director of Pakistan State Oil (PSO) and as CEO of Elengy Terminal Pakistan Limited (ETPL). At ETPL he spearheaded and commissioned the first 4.5 mtpa LNG import infrastructure for Pakistan in a world record 330 days — March 2015.

Source: Business Recorder, 29 December 2022 · <https://www.brecorder.com/news/40217243>